

Lecture 5

We have proved that $(\mathbb{Z}_n, +)$ is a group.

We also saw that $(\mathbb{Z}_n, \cdot)$ is never a group as

$[0]$ has no inverse under multiplication.

We will now find out which elements of $\mathbb{Z}_n$ have an inverse under multiplication.

Result: $[a] \in \mathbb{Z}_n$, $[a]$ has an inverse in $\mathbb{Z}_n$ under multiplication if and only if $\gcd(a, n) = 1$

Proof: Let $\gcd(a, n) = 1$.

Then we know $\exists x, y \in \mathbb{Z}$ such that

$$ax + ny = 1 \Rightarrow ny = ax - 1$$

$$\Rightarrow n \mid ax - 1.$$

$$\text{So } [ax] = [1]$$

$$\Rightarrow [a][x] = [1].$$

So $[x]$ is the inverse of $[a]$ in $\mathbb{Z}_n$.

Conversely let $[a]$ has an inverse in $\mathbb{Z}_n$ under multiplication say $[x]$.

$$\text{Then } [a][x] = [1]$$

$$\Rightarrow [ax] = [1]$$

$$\Rightarrow n \mid ax - 1 \Rightarrow ny = ax - 1$$

$$\Rightarrow ax + ny = 1.$$

& hence $\gcd(a, n) = 1$ as,

$\gcd(a, n)$ the least positive integer in

$$S = \{ ax' + ny' \geq 0 \mid x', y' \in \mathbb{Z} \}.$$

So $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(M_n(\mathbb{R}), +)$, $(\mathbb{Z}_n, +)$ are groups.

Since the binary operation is addition, these are called additive groups.

Similarly $(\mathbb{Q} \setminus \{0\}, \cdot)$, $(GL(n, \mathbb{R}), \cdot)$, $(\mathbb{R} \setminus \{0\})$ are called multiplicative groups.

For an additive group, we will denote $a * b$ by $a + b$

$\forall a, b \in G$. We will denote the identity element by 0 and the inverse of $a \in G$ by $(-a)$.

$(-a)$ is called the additive inverse of a .

In view of the associative law, we define

$$a + b + c = (a + b) + c = a + (b + c) \quad \forall a, b, c \in G.$$

for $m \in \mathbb{Z}$ and $a \in G$ we define

$$ma = \begin{cases} a + a + \dots + a & (m \text{ times}) \text{ if } m > 0 \\ 0 & \text{if } m = 0 \\ (-a) + \dots + (-a) & (|m| \text{ times}) \text{ if } m < 0. \end{cases}$$

From this we see that

$$(m+n)a = ma + na \quad \forall m, n \in \mathbb{Z}$$

$$(mn)a = m(na)$$

If G is a multiplicative group, then we will denote $a \star b$ by ab $\forall a, b \in G$. We will denote the identity of G by 1 and the inverse of $a \in G$ by a^{-1} .

(a^{-1}) is called the multiplicative inverse of a .

In view of associativity law, we define

$$abc = a \cdot (b \cdot c) = (a \cdot b) \cdot c \quad \forall a, b, c \in G.$$

further for $n \in \mathbb{Z}$ & $a \in G$, we define

$$a^n = \begin{cases} a \cdot a \cdots a & (n \text{ times}) \text{ if } n > 0 \\ 1 & \text{if } n = 0 \\ (a^{-1})(a^{-1}) \cdots (a^{-1}) & (-n \text{ times}) \text{ if } n < 0. \end{cases}$$

We see

$$a^{m+n} = a^m \cdot a^n$$

$$\forall m, n \in \mathbb{Z}, \quad a \in G.$$

$$(a^m)^n = a^{mn}$$

Theorem: Let G be a multiplicative group. Then

$$i) (a^{-1})^{-1} = a \quad \forall a \in G.$$

$$ii) (ab)^{-1} = b^{-1} \cdot a^{-1} \quad \forall a, b \in G.$$

Proof: i) We know that $a \cdot a^{-1} = 1 = a^{-1} \cdot a$.

So a is the inverse of (a^{-1}) .

$$\text{hence } (a^{-1})^{-1} = a$$

$$\text{ii) } (ab)(b^{-1}a^{-1}) = (a(bb^{-1}))a^{-1} = (a \cdot 1)a^{-1} = a \cdot a^{-1} = 1.$$

$$\& (b^{-1}a^{-1})(ab) = (b^{-1}(a^{-1}a)) \cdot b = (b^{-1} \cdot 1)b = b^{-1} \cdot b = 1.$$

$$\text{Hence } (ab)^{-1} = b^{-1}a^{-1}.$$

Let $(G, \cdot)$ be a multiplicative group. A subset $H \subseteq G$ is ~~closed~~ called closed under $\cdot$ in G if

$$a \cdot b \in H \quad \forall a, b \in H.$$

Ex: $\mathbb{Z}$ is closed in $(\mathbb{Q}, +)$.

$SL(n, \mathbb{R})$ is closed in $(\text{GL}(n, \mathbb{R}), \cdot)$.

Subgroups: A non-empty subset H of a group $(G, \cdot)$ is called a subgroup of G if

① H is closed under $\cdot$ in G .

② H is itself a group under $\cdot$.

Remark: Let e be the identity element in G .

Then $\{e\} \subseteq G$ & $G \subseteq G$. They are both subgroups of G . They are called trivial subgroups of G .

A subgroup H of G is called a proper subgroup of G if $H \neq \{e\}$ and $H \neq G$.

If H is a subgroup of G , then we write $H \leq G$.

Example: $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Q}, +)$

$(2\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$

$$2\mathbb{Z} = \{2n \mid n \in \mathbb{Z}\} = \text{set of even integers.}$$

$(SL(n, \mathbb{R}), \cdot)$ is a subgroup of $(GL(n, \mathbb{R}), \cdot)$

Lemma: Let G be a group. $H \leq G$. Then

i) The identity element 1_G of G is same as the identity element 1_H of H .

"ii) for each $a \in G$, $a^T \in H$.

Proof: (i) $1_G \cdot 1_H = 1_H$ as 1_G is the identity of G
 $\& 1_H \in G.$

$1_H \cdot 1_H = 1_H$ as 1_H is the identity of H .

$$\rightarrow \frac{1}{G} \cdot \frac{1}{H} = \frac{1}{H} \cdot \frac{1}{H}$$

$$\Rightarrow (I_G \ I_H) \cdot (I_H)^T = (I_H \ I_H) \cdot (I_H)^T \quad (\text{By } (I_H)^T \text{ both sides from right})$$

$$2) \quad I_G \cdot (I_H \cdot I_H^T) = I_H \cdot (I_H \cdot I_H^T)$$

$$\Rightarrow |G| \cdot |G| = |H| \cdot |G| = |H|$$

$$\Rightarrow I_G = I_H$$

(ii) Let a' be the inverse of a in H .

Need to show $a' = a^{-1}$

Since a^{-1} is the inverse of a in G , we have

$$a \cdot a^{-1} = 1_G = a^{-1} \cdot a$$

Since a' is the inverse of a in H , we have

$$a \cdot a' = 1_H = 1_G = a' \cdot a$$

$$\text{So } a \cdot a^{-1} = a \cdot a'$$

$$\Rightarrow a^{-1} \cdot (a \cdot a^{-1}) = a^{-1} \cdot (a \cdot a')$$

$$\Rightarrow 1_G \cdot a^{-1} = 1_G \cdot a'$$

$$\Rightarrow a^{-1} = a'$$

So $a^{-1} \in H$ $\forall a \in H$.

Theorem! (Subgroup Criterion) Let G be a multiplicative group. A non-empty subset H of G is a subgroup of G if and only if $ab^{-1} \in H$ $\forall a, b \in H$

(If G is an additive group, then a non-empty subset H of G is a subgroup if and only if $a - b = a + (-b) \in H$ $\forall a, b \in H$).

Proof: Let H is a subgroup of $(G, \cdot)$.

Let $a, b \in H$. Then $b^{-1} \in H$.

Since H is closed, $ab^{-1} \in H$.

Conversely, let $ab^{-1} \in H \quad \forall a, b \in H$.

We need to prove that H is a subgroup.

Since H is non-empty, $\exists h \in H$.

So $h, h^{-1} = 1 \in H$. ~~(a=b=h)~~ (By taking $a=b=h$).

By taking $a=1$ & $b=h$ we get

$$1 \cdot h^{-1} = h^{-1} \in H. \quad \forall h \in H.$$

To show that H is closed under $\cdot$,

we notice for $a, b \in H$, $a, b^{-1} \in H$.

So $a \cdot (b^{-1})^{-1} \in H \Rightarrow ab \in H. \quad \forall a, b \in H$.

Associative law holds in H as it holds in G .

Hence $H \leq G$.

Theorem: Let G be a group. H is a non-empty finite subset of G . Then

$H \leq G$ if and only if H is closed in G .

Proof: If $H \leq G$ then clearly H is closed in G .

Conversely, let H is closed in G .

We claim that H is a subgroup of G .

To show H is a subgroup of G , we need to prove $ab^{-1} \in H \quad \forall a, b \in H$ by previous theorem.

Consider $S = \{b^0, b^1, b^2, \dots, b^n, \dots\}$

Since H is closed in G , we have $S \subseteq H$.

Since H is finite, $\exists i \leq j$ such that

$$b^i = b^j$$

$$\Rightarrow (b^i)(b^i)^{-1} = (b^j)(b^i)^{-1}$$

$$\Rightarrow 1 = b^j \cdot b^{-i}$$

$$\Rightarrow b^{j-i} = 1 \quad \Rightarrow 1 = b^{j-i} \in H.$$

Also $b \cdot b^{j-i-1} = 1.$

$$\Rightarrow b^{-1} = b^{j-i-1} \in H.$$

Since H is closed, $ab^{-1} \in H \quad \forall a, b \in H$.

Hence H is a subgroup of G .